Initial system

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\dfrac{x_2}{10} + (-2\cos(t) - 1)\sin(x_1) + u \end{cases}$$

Series

$$x^0 = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \xi_0 + \begin{bmatrix} 1 \\ -\frac{1}{10} \end{bmatrix} \xi_1$$

Basis Lie elements and their coefficients

$$\mathcal{A}_1 : g_1 = \xi_0 \qquad v(g_1) = \begin{bmatrix} 0 & -1 \end{bmatrix}$$

$$\mathcal{A}_2 : g_2 = \xi_1 \qquad v(g_2) = \begin{bmatrix} 1 & -\frac{1}{10} \end{bmatrix}$$

Lie elements for principal part

$$\mathcal{A}_1 : \ell_1 = \xi_0$$

$$\mathcal{A}_2 : \ell_2 = \xi_1$$

The generating set of the Right ideal

Principal part

$$\mathcal{A}_1 : \tilde{\ell}_1 = \xi_0$$

$$\mathcal{A}_2 : \tilde{\ell}_2 = \xi_1$$

Approximating system

$$\begin{cases} \dot{x}_1 = -u \\ \dot{x}_2 = -tu \end{cases}$$

Stationary approximating system

$$\begin{cases} \dot{x}_1 = -u \\ \dot{x}_2 = -x_1 \end{cases}$$